# EMPATHY AND MINDFULNESS AS MODERATORS OF IMPLICIT BIAS EFFECTS ON THERAPEUTIC ALLIANCE

**Authors: Amna Maryam[1], Madison Wakely [1], Viktoria Bourmpoulia[1], and Dr. Sarah Bloch-Elkouby[2, 3]**
**1.Teachers College, Columbia University, 2. Yeshiva University, 3. Mount Sinai Hospital**

## Background

Therapeutic alliance is a key component of effective psychotherapy and treatment outcomes (Leonard et al., 2018). Previous research has looked at the interrelationships between empathy and mindfulness and therapeutic alliance. Empathy and mindfulness have been shown to enhance relational attunement, strengthen therapist presence, and facilitate rupture repair (Leonard et al., 2018; Baylar & Yurtsever, 2025). However, empathy and mindfulness have not been studied as moderators of implicit bias effects on therapeutic alliance.

## Current study

The present study addresses that gap by examining empathy and mindfulness as potential moderators in the relationship between implicit bias and therapeutic alliance.

**Hypotheses**

H1: Therapist empathy will moderate the relationship between implicit bias and therapeutic alliance.

H2: Therapist mindfulness will moderate the relationship between implicit bias and therapeutic alliance

## Methodology

The current research is a part of a larger study on the relationships between racial bias, clinical judgement and suicide risk assessment using virtual patients. There are two versions of Noah (virtual patient), black and white with which the clinicians interact with followed by a battery of questionnaires about their interaction. The current study used the following measures:

**1.Implicit Bias: Implicit Association Test** (IAT_d).

**2.Empathy: Therapist Empathy Scale** (TES; α = .98). (observer coded)

**3.Mindfulness: Mindful Attention Awareness Scale** (MAAS; α = .86).

**4.Alliance: 3RS Working Together subscale** (α = .84).

## Results



Black and white Noah

Preliminary correlational analysis (Pearson's R) revealed that mindfulness was significantly positively correlated with alliance (r = .16, p = .049). Empathy was positively but non significantly correlated with alliance (r = .14, p = .10). Bias was negatively but non significantly correlated with alliance (r = –.13, p = .12). Empathy and mindfulness were weakly correlated (r = .12, p = .19). Further moderation analyses were conducted using PROCESS (Hayes, Model 1 and Model 2).

- **Model 1 (Empathy as moderator)**: The overall model was not significant, $R^2$ = .04, F(3,116) = 1.49, p = .22. Neither bias (b = –.40, p = .13), empathy (b = .16, p = .14), nor their interaction (b = .19, p = .39) significantly predicted alliance.
- **Model 1 (Mindfulness as moderator):** The model approached significance, $R^2$ = .05, F(3,140) = 2.37, p = .07. Bias negatively predicted alliance (b = –.43, p = .07) and mindfulness positively predicted alliance (b = .30, p = .06), both at trend level. The interaction term was not significant (b = .43, p = .28).
- **Model 2 (Empathy + Mindfulness)**: The combined model was not significant, $R^2$ = .05, F(5,114) = 1.29, p = .27. Neither empathy nor mindfulness moderated the bias–alliance relationship.

Conditional effects probing suggested that bias predicted lower alliance particularly at low mindfulness, whereas at high mindfulness the bias–alliance relationship was attenuated. However, these trends were not statistically significant.

| Variable | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1. Alliance | — | | | |
| 2. Empathy | .14 | — | | |
| 3. Mindfulness | .16* | .12 | — | |
| 4. Implicit Bias | -.13 | .08 | .08 | — |

*Note.* $p < .05$.

## Conclusion

The present study tested whether therapist empathy and mindfulness moderated the impact of implicit bias on therapeutic alliance. Contrary to what was hypothesized, neither significantly buffered the bias–alliance link. However, mindfulness showed a consistent, trend-level positive association with alliance, suggesting its contribution may be more direct than interactive. Findings highlight mindfulness as a relational resource that enhances therapist presence and attunement, while underscoring the complexity of bias and the limitations of self-report empathy measures.

## Limitations

- At the time of submission, empathy coding was still in progress. Current findings are therefore preliminary and may shift once coding is complete.
- .Simulated interactions with virtual patients cannot fully capture the complexity and nuance of real clinical encounters.
- Reliance on self-report measures for mindfulness and alliance introduces the possibility of social desirability effects.
- The relatively small sample limits statistical power and generalizability of results.

## References

Baylar, E., & Yurtsever, S. S. (2025). Examining the Therapist Mindfulness in the Context of Therapeutic Alliance and Therapeutic Relationship: A Systematic Review. Psikiyatride Güncel Yaklaşımlar, 17(4), 687-706.

Leonard, H. D., Campbell, K., & Gonzalez, V. M. (2018). The relationships among clinician self-report of empathy, mindfulness, and therapeutic alliance. Mindfulness, 9(6), 1837-1844.

Thwaites, R., & Bennett-Levy, J. (2007). Conceptualizing empathy in cognitive behaviour therapy: Making the implicit explicit. Behavioural and Cognitive Psychotherapy, 35(5), 591-612.